Stock–recruitment conditioning and recruitment process error: ane.27.9aS

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Stock: Anchovy (*Engraulis encrasicolus*) in the southern part of Division 9.a (Gulf of Cadiz and southern part of Portugal), ane.27.9aS

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1 Recruitment conditioning for the projection

The stock–recruitment component of the OM was conditioned from the 100 historical realizations obtained in WD01. Each OM retains the SSB and recruitment trajectories corresponding to its SS3 bootstrap refit, with SSB determined in Q2 and age-0 recruitment entering in Q3 of the same year (ICES, 2024, 2025).

These trajectories were used to parameterize the stock–recruitment relationship used during 2025–2054 and to incorporate recruitment process error during the projection. Differences among OMs are represented through OM-specific stock–recruitment relationships, whereas annual variability within each trajectory is incorporated through recruitment process error.

2 Methods

2.1 Recruitment model diagnostics and specification

Recruitment structure was evaluated using the historical 1989–2024 series of Q2 SSB and Q3 recruitment from the same year.

2.1.1 Functional form of the stock–recruitment relationship

Beverton–Holt, Ricker, and segmented relationships were fitted to the same historical SSB–recruitment pairs using `FLSR` and `fmle`. Models were compared using log-likelihood, AIC, and ΔAIC, calculated relative to the lowest AIC among the candidate models. The fitted curves were also compared within the historical SSB range.

The functional form used in the reference OM was defined considering the model comparison and residual diagnostics.

2.1.2 Distribution and structure of residuals

Recruitment residuals were calculated on the logarithmic scale as

$$e_t = \log(R_t) - \log(\widehat{R}_t).$$

The residual distribution was characterized using the mean, standard deviation, skewness, kurtosis, and empirical quantiles. The assumption of normal error on the logarithmic scale was examined using a histogram, Q–Q plot, and the Shapiro–Wilk test (Thorson *et al.*, 2014; Punt *et al.*, 2016).

Temporal dependence was evaluated using ACF, PACF, Ljung–Box tests, an AR(1) model, and a runs test. The relationship between recruitment variability and SSB was examined using residuals and their absolute values against SSB and the Spearman correlation between $|e_t|$ and SSB.

Potential temporal changes in recruitment level were examined using changepoint and breakpoint analyses and the R/SSB series.

2.1.3 Stability of Beverton–Holt parameters

Sampling variation in the BH parameters was examined using 1,000 resamples with replacement of the historical SSB–recruitment pairs. For each resample, the model was refitted and a, b, log-likelihood, and AIC were recorded. This analysis was used as a diagnostic and did not introduce an additional source of parametric variability into the OM.

2.1.4 Specification retained for the reference OM

A Beverton–Holt relationship with lognormal recruitment deviations independent among years was used. The reference OM does not incorporate Ricker overcompensation, productivity regimes, biomass-dependent variance, environmental covariates, or an additional process for extreme events.

Variation in BH parameters among OMs arises from fitting the stock–recruitment relationship to the historical trajectories of each SS3 refit.

2.2 OM-specific stock–recruitment relationships

For each of the 100 OMs, Q2 SSB and Q3 recruitment were extracted for 1989–2024, yielding 36 historical SSB–recruitment pairs per OM.

For each OM, a Beverton–Holt (BH) relationship was fitted:

$$\widehat{R}_{t,i} = \frac{a_i SSB_{t,i}}{b_i + SSB_{t,i}},$$

where i identifies the OM, t the year, a_i the asymptotic recruitment, and b_i the SSB at which expected recruitment equals $a_i/2$.

An unrestricted fit was initially obtained using `FLSR` and `fmle` and used as a diagnostic. For OM parameterization, a_i and b_i were constrained to positive values through a logarithmic transformation and estimated by minimizing

$$Q_i = \sum_t \left[\log R_{t,i} - \log \widehat{R}_{t,i}\right]^2,$$

using the Nelder–Mead algorithm implemented in `optim`, with `maxit = 10000` and `reltol = 1e-12`.

For each OM, historical recruitment variability around the BH relationship was estimated as

$$\sigma_{R,i} = \mathrm{SD}_t \left[\log R_{t,i} - \log \widehat{R}_{t,i}\right].$$

A convergence code of zero and finite values of a_i, b_i, and $\sigma_{R,i}$ were required to retain the fit. Positive values of b_i, regardless of their magnitude, were not used as an additional exclusion criterion. ## Construction of the stock–recruitment component in `FLSRsim`

The a_i and b_i parameters estimated for each OM were incorporated into an `FLSRsim` object within the FLBEIA simulation structure (Garcia *et al.*, 2017). The correspondence between each OM and its bootstrap refit was maintained during this assignment.

The model was specified as `model = "bevholt"`, with a_i and b_i constant throughout the projection period for each OM. The seasonal structure retains SSB in Q2 and recruitment entry in Q3. The recruitment proportion was set to one in Q3 and zero in the remaining quarters, with `timelag["year", ] = 0` and `timelag["season", ] = 2`.

At this stage, `uncertainty` multipliers were set to one, so no process error was incorporated during the projection. Recruitment process error is subsequently incorporated into this structure.

2.3 Expansion from 100 OMs to 1,000 trajectories

Each of the 100 OMs was replicated 10 times, generating a set of 1,000 trajectories. The 10 replicates of each OM retain the same biological state and the same stock–recruitment parameters a_i, b_i, and $\sigma_{R,i}$ as the source OM.

Replication does not introduce an additional source of variability. Differences among the 10 replicates of the same OM are subsequently generated through annual recruitment process error during the projection.

2.4 Recruitment process error during the projection

Recruitment process error was incorporated annually during 2025–2054. Two parameterizations of the standard deviation of recruitment residuals were considered.

In the reference configuration, a common value of σ_R was used, defined as the median of the values estimated for the 100 OMs:

$$\sigma_R^{\text{ref}} = \text{median}_{i=1,\ldots,100}\left(\sigma_{R,i}\right).$$

In the sensitivity configuration, each trajectory retained the σ_R value estimated for its source OM:

$$\sigma_{R,j}^{\text{OM}} = \sigma_{R,\text{OM}(j)},$$

where j identifies one of the 1,000 trajectories. Therefore, the 10 replicates of the same OM share the same σ_R value.

For each year and trajectory, a standard normal deviate was generated:

$$z_{t,j} \sim N(0,1).$$

The same matrix of $z_{t,j}$ values was used in both configurations. For each parameterization s, the error on the logarithmic scale was calculated as

$$\varepsilon_{t,j}^{(s)} = z_{t,j}\sigma_{R,j}^{(s)},$$

and the multiplier applied to recruitment as

$$m_{t,j}^{(s)} = \exp\left[\varepsilon_{t,j}^{(s)} - \frac{1}{2}\left(\sigma_{R,j}^{(s)}\right)^2\right].$$

The term $-\sigma_R^2/2$ corresponds to the bias correction for the lognormal distribution, such that

$$E\left[m_{t,j}^{(s)}\right] = 1$$

(Maunder *et al.*, 2006).

The multipliers were applied exclusively to Q3 recruitment during 2025–2054. In Q1, Q2, and Q4, multipliers were kept equal to one.

Thus, differences among the 100 OMs arise from their historical trajectories and the parameters of their stock–recruitment relationships, whereas differences among the 10 replicates of the same OM arise from annual recruitment process error during the projection (Punt *et al.*, 2016).

2.5 Implementation checks

Checks verified that: (i) all 100 BH fits converged and had finite parameters; (ii) the correspondence among each OM, its bootstrap refit, and its stock–recruitment parameters was maintained; (iii) each OM had 10 replicates; (iv) recruitment multipliers were positive and finite; and (v) these multipliers were applied exclusively in Q3 during 2025–2054.

For the reference configuration, it was additionally verified that the realized standard deviation of the logarithmic deviations differed by less than 0.01 from the σ_R^{ref} value used to generate them.

For the OM-specific configuration, realized standard deviations were calculated by trajectory and retained as a diagnostic. No acceptance criterion based on exact correspondence with $\sigma_{R,i}$ was applied, because each trajectory contains 30 annual realizations.

3 Results

3.1 Model diagnostics and retained specification

The comparison of Beverton–Holt, Ricker, and segmented relationships is presented in Table 1. Models were compared using log-likelihood, AIC, and ΔAIC. Figure 1 shows the fitted relationships within the historical SSB range.

Table 1: Comparison of Beverton–Holt, Ricker, and segmented formulations fitted to the historical Q2 SSB–Q3 recruitment series. Delta AIC is calculated relative to the lowest AIC among the candidate models.

Model	logLik	AIC	Delta AIC
Segmented	36.457	-68.915	0.000
Beverton-Holt	36.419	-68.837	0.077
Ricker	35.915	-67.830	1.085

3.1.1 Residuals and process error

The distribution of logarithmic recruitment residuals is summarized in Table 2. Figure 2 shows the residual time series, their distribution, the Q–Q plot, and their relationship with SSB.

Temporal dependence is summarized in Table 3, including the AR(1) coefficient and the runs test.

3.1.2 Parameter stability

Of the 1,000 resamples of the historical SSB–recruitment pairs, 840 produced fits (84.0%). The distribution of the a and b parameters obtained from these fits is summarized in Table 4.

Table 5 summarizes the diagnostics and the specification retained for recruitment in the reference OM.

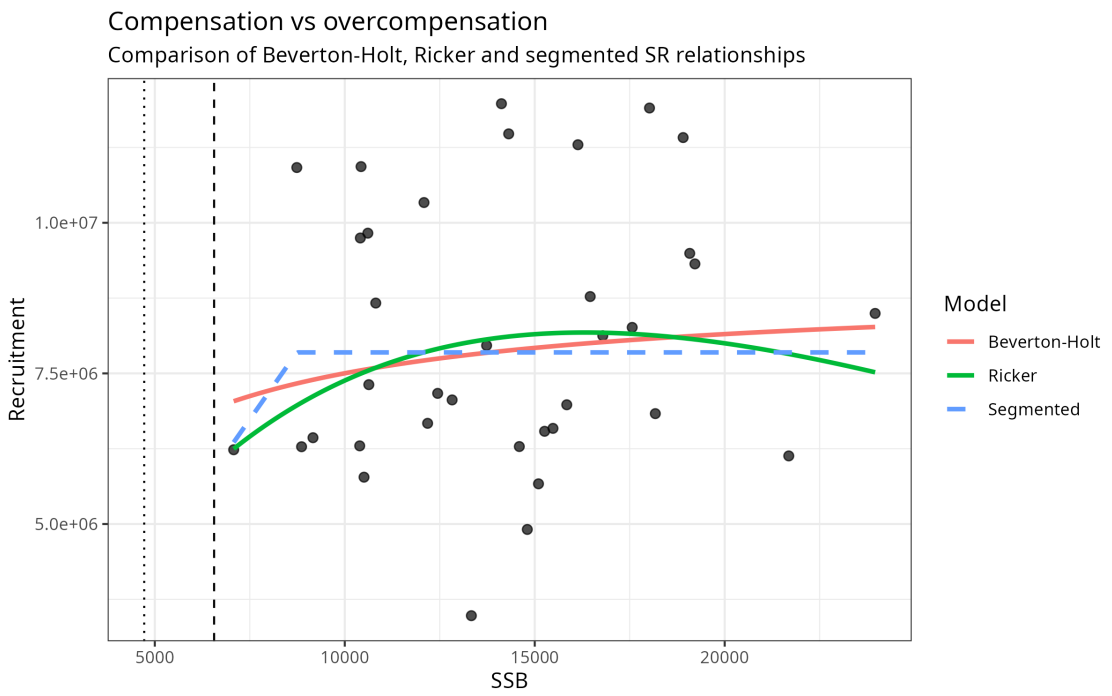


Figure 1: Comparison of Beverton–Holt, Ricker, and segmented stock–recruitment relationships fitted to Q2 SSB and Q3 recruitment during 1989–2024. Vertical lines indicate Blim and Bpa.

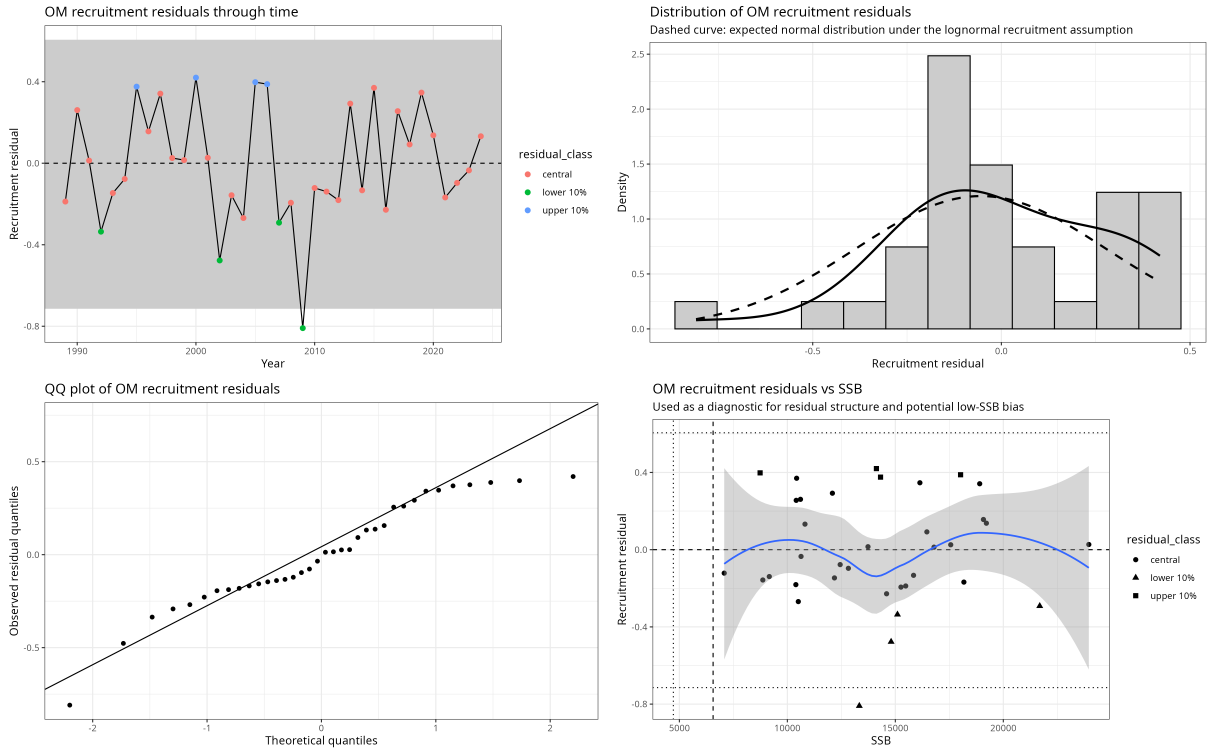


Figure 2: Diagnostics of logarithmic recruitment residuals used to evaluate the process-error assumption: time series, distribution, Q–Q plot, and residuals against SSB.

Table 2: Summary of logarithmic recruitment residuals from the reference fit.

Diagnostico	Valor
Years	36
Mean residual	-1.00112212953165e-05
Residual SD	0.279970821569764
sigmaR used	0.33
sigmaR source	SS3
Skewness	-0.408162185048044
Kurtosis	3.06648276515079
P05	-0.371102688830716
Median	-0.0109648125497985
P95	0.390561727335415
Minimum	-0.809194536778948
Maximum	0.420079026315124
Proportion within ±1.96 sigmaR	0.972222222222222

Table 3: Summary of temporal-dependence diagnostics for recruitment residuals.

Diagnostic	Estimate	p-value
AR(1), phi	0.1171	0.4771
AR(1) intercept	-0.0002	0.9967
Runs test	-1.6910	0.0908

Table 4: Bootstrap distribution of Beverton–Holt parameters in successful fits.

Parametro	P05	P25	Median	P75	P95
a (individuals)	7782975.08	NA	9305973.94	NA	12396209.1
b (t)	288.68	1312.17	2516.81	4079.62	7692.3

Table 5: Summary of recruitment diagnostics and decisions retained for the reference OM.

Component	Diagnostic	Reference OM
Should a stock-recruitment relationship be used?	Q2 SSB linked to Q3 recruitment	Yes
Which SR relationship should be used in the base OM?	BH vs segmented vs Ricker comparison using Q2 SSB and Q3 recruitment	Beverton-Holt
Should Ricker overcompensation be used?	Ricker comparison and recruitment at high Q2 SSB	No
Should recruitment deviations be lognormal?	Q3 recruitment residual distribution, QQ plot, Shapiro-Wilk and tail diagnostics	Yes
Should recruitment deviations be autocorrelated?	ACF, PACF, Ljung-Box, AR(1), runs test on Q3 recruitment residuals	No
Should productivity regimes be included?	Changepoint, breakpoints and productivity diagnostics using Q3 recruitment	No
Should biomass-dependent recruitment variance be included?	Absolute Q3 recruitment residuals vs Q2 SSB, biomass-class tests	No
Should environmental covariates be included?	No tested mechanistic environmental driver	No
Should maternal or spawner-age effects be included?	Q3 recruitment residuals vs Q2 spawner age structure	No, unless diagnostics show a clear relationship

Table 5: Summary of recruitment diagnostics and decisions retained for the reference OM. *(continued)*

Component	Diagnostic	Reference OM
Should sporadic recruitment events be included?	Extreme Q3 residual years and Q3 cohort strength classification	No as explicit process; retained through stochastic recruitment variability
Should SR parameter uncertainty be explored?	Bootstrap BH parameters using Q2 SSB and Q3 recruitment	Not in base OM; use sensitivity scenarios

3.2 Parameterization of stock–recruitment relationships

The 100 OM-specific BH fits had a convergence code of zero. The median a was 9,069,713 individuals and the median b was 2,815.8 t. In 7 OMs, $b < 1$ t was obtained.

The unrestricted diagnostic fits produced 0 values of $b \leq 0$. The positive parameterization used in the OM retained these OMs through the constraint $b_i > 0$.

The distribution of a, b, and σ_R among the 100 OMs is presented in Table 6. Figure 3 shows the 100 BH relationships used during the projection.

Table 6: Distribution of Beverton–Holt parameters and the historical standard deviation of recruitment residuals across the 100 conditioned OMs.

Parametro	P05	Median	P95
a (individuals)	7269905.357	9069712.736	11663725.338
b (t)	0.000	2815.795	6115.389
sigmaR	0.184	0.216	0.254

3.3 Historical recruitment variability

The σ_R values estimated for the 100 OMs ranged from 0.1679 to 0.2782, with a median of 0.2163 (Table 7).

Table 7: Distribution of historical standard deviations of logarithmic recruitment residuals across the 100 OMs.

n OMs	Minimum	P05	Median	Mean	P95	Maximum
100	0.1679	0.1844	0.2163	0.2158	0.254	0.2782

3.4 Recruitment process error during the projection

The reference configuration used $\sigma_R = 0.2163$ in all 1,000 trajectories. In the OM-specific configuration, assigned values ranged from 0.1679 to 0.2782.

The realized mean multipliers were 1.000083 and 1.000075, respectively (Table 8). Their distributions are shown in Figure 4.

Figure 4 compares the distributions of the multipliers generated under both parameterizations. Both configurations use the same matrix of standard normal deviates; differences between the distributions result from the σ_R values used to scale the deviations.

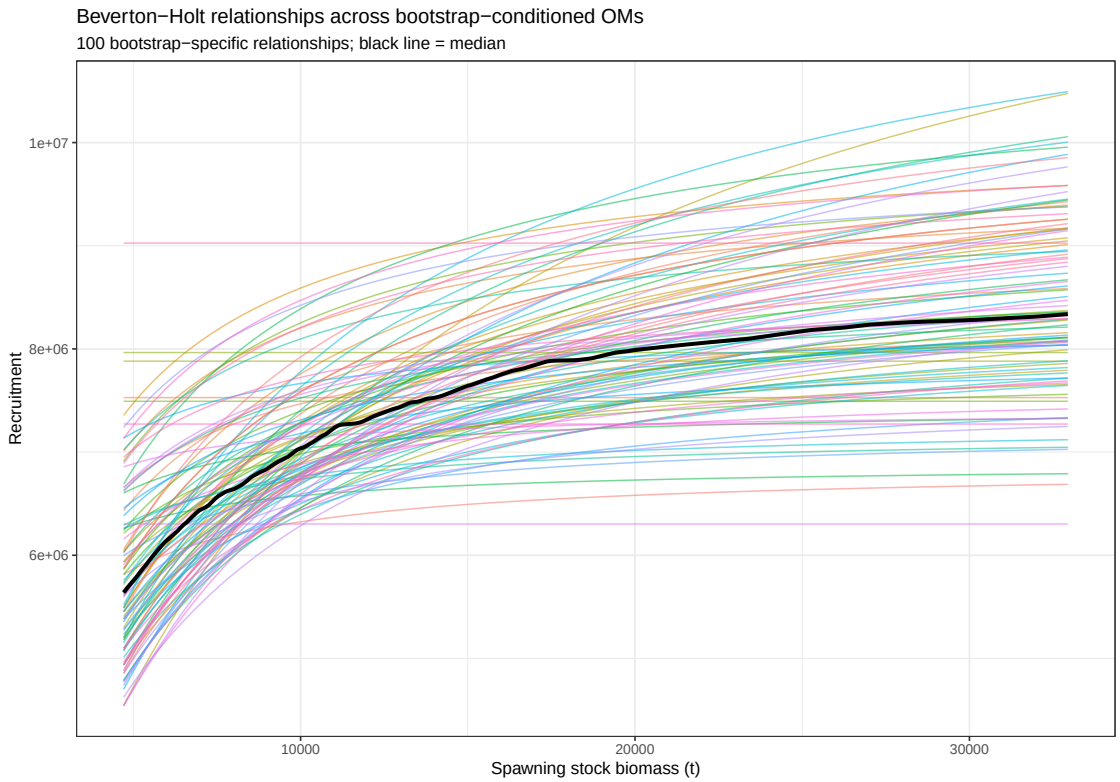


Figure 3: Beverton–Holt relationships used to parameterize the stock–recruitment component of the 100 conditioned OMs. Lines represent OM-specific relationships, and the black line corresponds to the pointwise median over the historical SSB range.

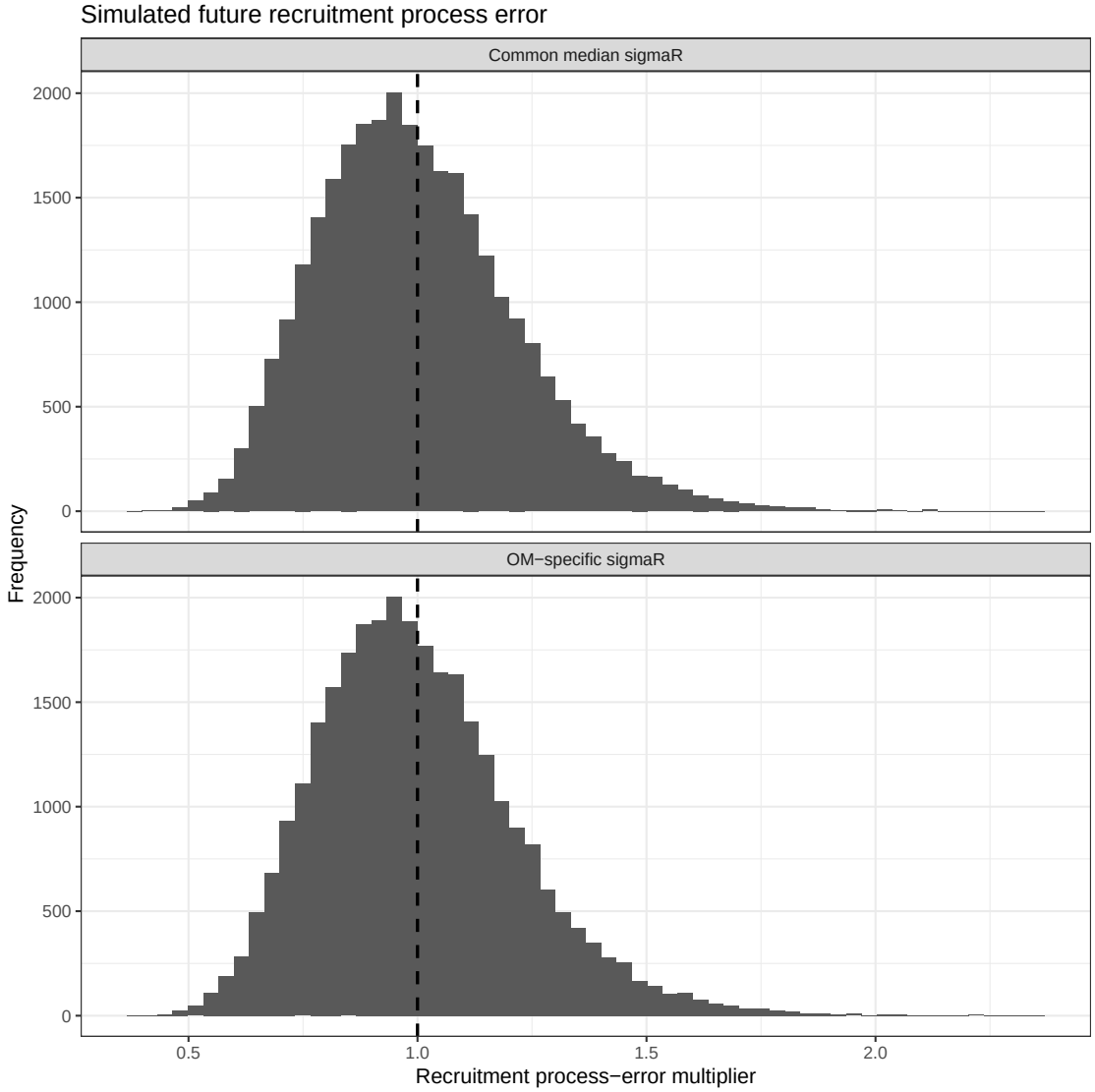


Figure 4: Distribution of recruitment process-error multipliers during 2025–2054 for the reference and OM-specific configurations. Both configurations use the same standard normal deviates for the 1,000 trajectories.

Table 8: Parameterization and summary of recruitment process-error multipliers for the reference configuration and the OM-specific configuration during 2025–2054.

Configuration	n	sigmaR min.	sigmaR median	sigmaR max.	Mean multiplier	Median multiplier
Common median	1000	0.2163	0.2163	0.2163	1.0001	0.9771
OM-specific	1000	0.1679	0.2163	0.2782	1.0001	0.9774

4 Reproducibility

The construction sequence for this component is `05b_recruitment_model_diagnostics.R` → `06_fit_SR_bootstrap.R` → `07_build_SR_reference.R` → `08_expand_OM_100_to_1000.R` → `09_recruitment_process_er`
The scripts start from the 100 OMs generated during historical conditioning and produce the SR objects subsequently used in the OM projections.

`06_fit_SR_bootstrap.R` generates the BH fits and σ_R estimates; `07_build_SR_reference.R` constructs the `FLSRsim` for the 100 OMs; `08_expand_OM_100_to_1000.R` expands the objects to 1,000 trajectories; and `09_recruitment_process_error_with_robustness.R` generates and assigns the future recruitment multipliers. The seed used for process error is `1234`.

The final objects are `SRs_reference_1000iter_process_error.rds` and `SRs_om_specific_1000iter_process_error.rds`. Deviations, multipliers, and the shared matrix of standard normal deviates are stored in `data/Rdata`, whereas the tables and figures used by this WD are stored in `outputs/recruitment/tables` and `outputs/recruitment/figures`.

5 Conclusions

- The 100 OMs were parameterized using OM-specific Beverton–Holt relationships based on Q2 SSB and Q3 recruitment during 1989–2024.
- Each OM retains the a_i, b_i, and $\sigma_{R,i}$ parameters corresponding to its SS3 bootstrap refit.
- Each OM was replicated 10 times to generate 1,000 trajectories. This replication does not introduce additional variability.
- The reference configuration uses a common σ_R value, defined as the median of the values estimated for the 100 OMs. The sensitivity configuration retains the OM-specific σ_R.
- Process error is applied annually to Q3 recruitment during 2025–2054, using the same standard normal deviates in both configurations.

References

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